



India Precision Engineering

Sector Thesis & Deep-Dive

\$165Bn

Global TAM (2025)

6.8%

Global Growth CAGR (2025-33)

\$8Bn

India TAM (2025)

9.5%

Indian Growth CAGR (2025-33)

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From market mechanics to investment positioning.

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Sector Definition & Global Market

Technology, processes and where India fits in the global precision engineering landscape

What is Precision Engineering?

Definition	The design and manufacture of components to extremely tight accuracy standards, ensuring consistent performance in high-reliability industries such as aerospace, automotive, electronics, and medical devices.
Tolerances	Parts machined to ± 0.005 - 0.05 mm for standard aerospace, sub-micron for optics and semiconductor tools. Even a few micron deviation can cause catastrophic failure in critical applications.
Materials	High-grade alloys: Titanium, Inconel, Stainless steel, Aluminum, CMC composites. Material selection is critical as each behaves differently under machining loads, temperature, and stress cycles.
Verification	Coordinate Measuring Machines (CMMs), laser interferometers, optical comparators, X-ray CT. Every critical part traceable to national measurement standards. Quality is built in, not inspected in.

Key End-Use Sectors

- Aerospace & Defence**
Engine components, airframes, landing gear, guided munitions, radar systems
- Automotive & EV**
Engine blocks, transmission gears, EV motor housings, drivetrain components
- Medical Devices**
Orthopaedic implants, surgical instruments, stents, diagnostic imaging equipment
- Electronics & Semiconductors**
PCB housings, micro-connectors, semiconductor fab tooling, precision optics
- Energy & Nuclear**
Wind turbine gearboxes, nuclear reactor vessels, steam generator components
- Industrial Machinery**
Pumps, valves, compressors, hydraulic actuators, precision robotics systems

Technology Stack & Manufacturing Processes

01

Multi-Axis CNC Machining

Simultaneous 5+ axis control produces complex geometries in one setup. Vertical and horizontal machining centers manufacture aerospace brackets, engine housing, and transmission components, reducing setup time ~60% vs 3-axis. India operates 8,000+ CNC units.

02

Electrical Discharge Machining (EDM)

Controlled electrical discharges erode material without mechanical contact. Essential for hardened steels and complex internal profiles – die cavities, turbine cooling holes, gear teeth. Zero cutting forces means no distortion of fragile or complex workpieces.

03

Precision Grinding & Finishing

Surface, cylindrical and centreless grinding refines dimensions to sub-micron level, critical for bearing raceways, crankshaft journals, valve bores. Optical polishing reaches nanometer-scale for defence and semiconductor optics.

04

Precision Forging & Forming

Closed-die forging produces near-net-shape parts with superior grain flow and fatigue resistance vs. machined billets. Critical for aerospace bulkheads, automotive crankshafts, and rail axles.

05

Metal Additive Manufacturing

Direct Metal Laser Sintering and Selective Laser Melting build complex lattice structures and consolidated assemblies impossible by regular machines conventionally. Growth in aerospace and medical implants. India's aerospace sector is in first-use programme development.

06

Metrology & Quality Systems

Laser scanning of complex surfaces, X-ray CT for internal features, ultrasonic NDT for hidden flaws. Every critical aerospace part has a dimensional report. NADCAP accredits special processes, each requiring 200+ verified quality requirements per audit cycle.

Global Precision Engineering: Market Size & Dynamics

\$123.5Bn

Global Precision Machining

→ \$228Bn by 2033 · 8.1% CAGR

\$40Bn

Global Precision Tooling

→ \$72Bn by 2033 · 7.6% CAGR

\$30Bn+

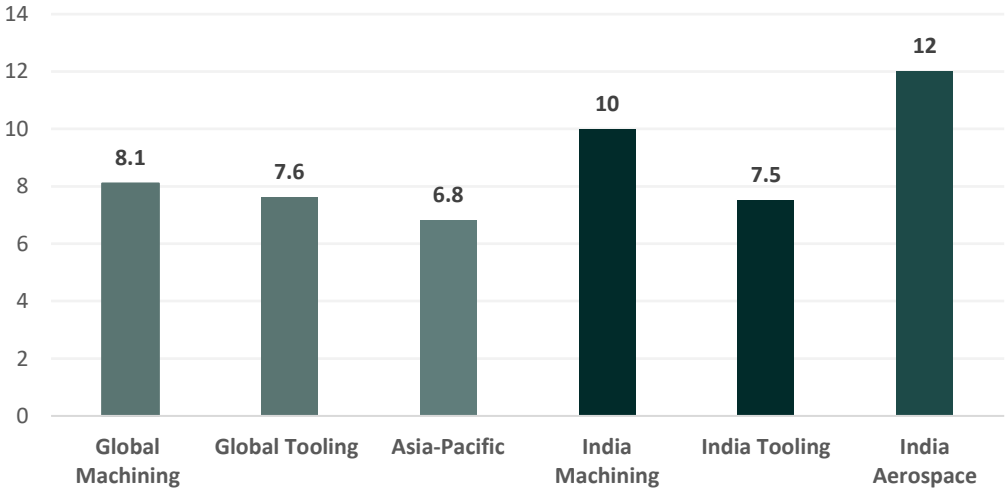
Global Aerospace
Precision Machining Size

India's share growing

6.8%

Asia-Pacific CAGR

Fastest growing region



CAGR Comparison – India vs Global Benchmarks (%)

Germany / Japan / USA

Dominate ultra-precision: semiconductor lithography, aerospace engine cores. Decades of process IP. Premium pricing power. Not easily displaced in highest tiers.

China

Rapidly built mid-tier capacity. Still imports ~80% of high-end CNC machine tools. Quality reliability concerns driving OEM diversification.

Taiwan / South Korea

Dominant in electronics and semiconductor precision hardware. Dense supplier ecosystems. Rising labour costs reducing competitiveness in mid-complexity work.

India → The Opportunity

Competitive in mid-to-high complexity range. Skilled-labour costs 20-30% below European equivalents. Improving quality infrastructure. IP environment differentiate vs. China.

Indian Market Size & Demand Drivers

From an \$8Bn precision equipment market today to an \$18-22Bn total sector opportunity

India Precision Engineering: Market Size & Trajectory

\$18-22Bn

India's Total Addressable Market in Precision Engineering
2023 estimate

\$6.6Bn

Precision Machining 2025

→ \$14 Bn by FY2033 at 10% CAGR

₹1.5 Bn

Precision Tools in 2025

→ \$2.7Bn by 2033 at 7.5% CAGR

>\$5B

Component Exports by 2030

Fasteners, bearings, aerospace parts

8,000+

Precision Machining Units
in India

85% of national machine-tool
output is CNC

End-Use Sector	Share	Growth	Key Driver	Investment Note
Automotive & Auto Components	~52%	Moderate, EV adds complexity	ICE→EV mix shift; exports growing	Lower multiple, EV re-rating opportunity
Aerospace & Defence	~20%	High ~15%+ CAGR	Indigenization, Boeing/Airbus sourcing	Highest valuation premium
Electronics & Semiconductors	~10%	High	PLI schemes, iPhone manufacturing	Growing, low base effects compound
Industrial Machinery & Energy	~10%	Moderate-High	Railways capex, renewables	Steady, underappreciated by market
Medical Devices	~5%	High (low base)	Make in India, ageing population	Nascent, watch for inflection signal
Others (Nuclear, Rail, etc.)	~3%	Moderate	Nuclear expansion, metro rail	Niche but durable, policy-backed

Eight Structural Demand Drivers: 1 of 2

These are not cyclical tailwinds, they are multi-year structural shift operating independently of short-term macroeconomic variability.

01

Defence Indigenization

India's defence production reached ₹1.27 lakh crore in FY2023-24, a 174% increase since FY2014 and an all-time record high. 65% of equipment is now domestically produced (up from ~30% a decade ago). The private sector contributes 21% and rising. Government target: ₹3 lakh crore by FY2029, implying 20%+ CAGR. India has ~16,000 defence-focused MSMEs in the supply chain.

02

Aerospace Build-Out: Civil & Military

India becomes world's third-largest aviation market by 2030. HAL order book: ₹1,89,300 crore, FY25 revenue: ₹30,981 crore. Tejas Mk2, AMCA and Prachand helicopter programmes represent decades of domestic demand. Boeing now sources \$1B+/yr from India (from \$250M a decade ago); Airbus targets \$2B by 2030.

03

Electric Vehicle Transition

Record 5.36M vehicle exports in FY2025. EVs require a more precision-intensive component set than ICE: permanent magnet motors, helical reduction gearboxes (tight tolerances), power electronics thermal management housings, and high-voltage connector assemblies. PLI incentives for EV components subsidise localisation. Companies qualifying early face minimal competition.

04

China+1 Supply Chain Realignment

US tariffs of 25% on Chinese manufactured goods, post-COVID fragility, and corporate ESG commitments are driving structural reconfiguration of global supply chains. India is the primary beneficiary: 20-30% lower skilled-labour costs vs Europe, improving quality infrastructure, stronger IP protection. This is structural repositioning, not a temporary trade diversion.

Eight Structural Demand Drivers: 2 of 2

05

Make in India & PLI Schemes

PLI schemes across electronics, auto components, aerospace, advanced chemistry and semiconductors channel ₹2.5-3 lakh crore into domestic manufacturing. Performance-linked tied to actual output, creating durable capacity not short-term subsidy dependence. As PLI beneficiaries expand production, they create multiplier pull for precision tooling and machining. Aerospace & Precision Engineering SEZ under development in Telangana.

06

Electronics & Semiconductor Manufacturing

Electronics value-addition rose from ~30% to ~70%; target 90% by FY2027. Apple's Tata iPhone programme requires tens of thousands of precision-machined metal housings and thermal components annually. India's semiconductor mission, Tata Electronics and foreign foundry partners will create demand for precision components used in fab equipment, a market exceeding \$100B globally.

07

Renewables & Nuclear Expansion

India's 500 GW renewable target by 2030 requires wind turbine gearboxes, generator shafts, solar tracker actuators, all precision-engineered. Wind is growing 30%+ annually. Nuclear expansion (12+ reactors planned) demands some of the most demanding precision fabrication in the industry. These are long-duration, high-margin programmes.

08

Infrastructure Capex Cycle

Railways received ₹2.78 lakh crore in Union Budget 2026, the largest ever. Vande Bharat trainsets, 27-city metro systems, and dedicated freight corridor wagons require precision-engineered bogies, wheelsets, brake systems, and pneumatic actuators. India's broader infrastructure capex of roads, ports, airports drives demand for precision construction equipment and hydraulic components.

Value Chain & Industry Economics

Where value is created, who captures it, and what each business model looks like

The Precision Engineering Value Chain: Layer by Layer

01 Raw Materials

Tata Steel · JSW · SAIL · Hindalco · Imported specialty alloys

Standard steel and aluminium adequately supplied domestically. Specialty alloys (titanium, Inconel) still largely imported, a structural vulnerability.
Commodity economics, limited differentiation.

10–20%

EBITDA

Lowest

02 Machine Tools & Industrial Tooling

Lakshmi Machine Works · Jyoti CNC · Kennametal

The 'Mother Industry'. Technology IP, recurring tooling consumable demand, and aftermarket servicing drive highest upstream margins. India is upgrading from 3-axis to 5-axis is a secular capex cycle.

15–30%

EBITDA

High

03 Precision Component Manufacturing

Sansera Engineering · Bharat Forge · Sundram Fasteners

CNC machining, forging, casting as Tier-1/2 OEM suppliers. Scale-driven economics. OEM pricing power tempers margins. Mix shift toward aerospace is the primary margin-expansion lever.

12–20%

EBITDA

Moderate

04 High-Complexity Precision Systems

MTAR Technologies · Dynamatic Technologies · Azad Engineering

Aerospace tooling, engine handling systems, electro-mechanical assemblies. NADCAP certification + 5-10 year qualification cycles + 20-40 year programme lifetimes create near-permanent moats.

20–35%

EBITDA

Highest

05 OEM Integration

HAL · BEL · Tata Motors · Airbus India · Boeing India

System integration and programme management. OEMs exercise pricing power downward. Brand, customer relationships, and qualification authority are the assets.

10–15%

EBITDA

Moderate

Value capture increases significantly as companies move from commodity machining toward certified, programme-embedded precision systems

Revenue Models & Margin Architecture

Business model is the single most important determinant of economics. The four models below represent a full spectrum from commodity contract machining to long-duration programme supply.

Contract Manufacturing

Volume × machining hours × complexity

EBITDA
12-18%
Cycle
Annual
Moat
Low
Visibility
Low

Revenue driven by machining hours multiplied by component complexity rate. Meaningful volume sensitivity, a 10% volume drop causes 20-30% EBITDA reduction. Requires continuous capex reinvestment. Working capital intensive.

Build-to-Print Manufacturing

Customer-owned design; supplier executes

EBITDA
14-20%
Cycle
Programme
Moat
Medium
Visibility
Medium

Customer provides drawing, manufacturer develops process and tooling. Common in aerospace Tier-2 structural machining. Qualification stickiness creates medium moat. Price pressure at programme renewal points. Design ownership remains with OEM.

Long-Term Supply Agreements

Programme lifecycle contracts (20-30 years)

EBITDA
18-28%
Cycle
20-30 yrs
Moat
High
Visibility
High

Multi-year contracts tied to specific programmes. Revenue visibility spans product lifecycles. MTAR/ISRO and Dynamatic/Boeing exemplify this model. Order book/revenue ratios of 4-8x are common.

Value-Added Engineering

Design ownership + sub-system assembly

EBITDA
25-35%
Cycle
30+ yrs
Moat
Very High
Visibility
Very High

Highest-margin model. Companies own design, produce complete sub-systems (landing gear assemblies, guided munition sub-systems) and manage the supply chain. Integrated engineering increases pricing power and switching costs to near-infinite levels.

— 04

Structural Moats

Why the best companies stay best: six competitive barriers

Six Structural Competitive Moats

Moats here are physical, process-embedded and regulatory, not digital. They are expensive to build, slow to develop, and practically impossible to undercut on price alone.

01 Certification & Qualification Lock-In

AS9100D, NADCAP (heat treatment, coatings, NDT, welding, chemical processing), ISO 13485, each requiring 18-36 months of documented process control, first-article inspection, PPAP, and continuous audit cycles with 200+ verified requirements. Once achieved, certifications are embedded in OEM supply agreements. An uncertified competitor cannot enter an aerospace supply chain regardless of price.

03 Process IP in Difficult Materials

Machining titanium, Inconel, or ceramic matrix composites requires years of parameter optimisation: cutting speed, tool geometry, coolant strategy, fixture design, and thermal distortion control. MTAR Technologies' cryogenic engine expertise took over a decade to develop. Azad Engineering's multi-NADCAP accreditations reflect years of special-process investment. This process IP is non-codified, non-transferable, and inseparable from the workforce that built it.

05 Scale & Machine Utilisation Economics

Precision machining is high-fixed-cost: a 5-axis machining centre costs ₹2-8 crore; a NADCAP-accredited heat treatment furnace ₹5-15 crore. Incremental revenue above ~65% utilisation flows disproportionately to EBITDA (operating leverage). Scale also enables higher-capability machines (5-axis vs 3-axis) and more comprehensive quality systems, creating a compounding advantage smaller competitors cannot match.

02 Programme-Embedded Supplier Relationships

Aerospace and defence supply agreements are programme-specific, not annual. When Dynamatic wins landing-gear hub supply for HAL Dhruv or Airbus A320neo, that relationship runs coextensive with the programme's production life of 20-40 years. OEM switching a supplier mid-programme requires financing a new qualification journey costing \$2-10M and 3-5 years. This mutual sunk cost creates bilateral lock-in.

04 Vertical Integration Creates Physical Stickiness

Companies controlling multiple value chain stages: forging, precision machining, heat treatment, surface coating, assembly, and inspection offer OEM customers superior cost control, faster development, and simpler supply-chain management. Aequs Aerospace operates 1.66 million CNC-hours per year with in-house forging and Boeing-approved surface treatment in one SEZ. This integration reduces OEM supply-chain risk and creates switching costs through physical asset co-location.

06 Two-Sided Qualification Investment

Both supplier and OEM invest substantially: supplier develops processes, fixtures, SPC systems, and documentation; OEM invests in supplier audits, PPAP review, and first-article inspection. These mutual sunk costs create two-sided lock-in. A supplier embedded in 8-10 active programmes with major OEMs has built a qualification portfolio that would take a new entrant 15-20 years to replicate.

China+1 Opportunity

A structural repositioning of global supply chains, not a one-year trade diversion.
India is the primary beneficiary.

Boeing India Sourcing	\$1B+ / year (from \$250M a decade ago)
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Airbus India Target	\$2B / year by 2030
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Precision Exports (FY2030 proj.)	>\$5 billion
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US Tariff on China Mfg.	25-50% — structural realignment driver
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India's Five Competitive Advantages

Labour Cost

Cost

Skilled CNC operator labour ~20-30% of European equivalents. Comparable to China for mid-complexity precision work. Durable cost advantage in machining-intensive products where setup and operation time dominates cost.

Quality Infrastructure

Quality

1,000+ AS9100-certified precision facilities. NADCAP accreditations growing. ISO-calibrated metrology labs. Verifiable, auditable quality systems meeting global OEM requirements for aerospace and defence supply chain entry.

Language & IP Protection

Strategic

English proficiency enables direct OEM engineering collaboration, no translation overhead in technical specs, drawing interpretation, or quality documentation. Stronger IP protection than China reduces technology transfer risk significantly.

Engineering Talent Pipeline

Talent

1.5M+ engineering graduates annually. Growing pool addressable for GD&T, 5-axis CNC programming, and precision metrology skills. Technical universities producing increasingly industry-ready graduates.

Logistics Infrastructure

Logistics

Dedicated freight corridors operational. 33 ports. GatiShakti master plan reducing transit times and costs. Precision clusters in Pune, Bengaluru, Chennai, Coimbatore accessible to both domestic OEMs and export shipping lanes.

Key Players & Competitive Landscape

From large-cap OEMs to high-complexity specialists: who's who and where they sit

Key Listed Companies: Full Landscape Overview

Company	Revenue	Sector Focus	Complexity	EBITDA	Certification
HAL	₹30,981 Cr	Aerospace OEM	Very High	31-34%	AS9100, CEMILAC
BEL	₹23,769 Cr	Defence Electronics	Very High	27-29%	AS9100, ISO
Bharat Forge	₹15,123 Cr	Forgings — Auto/Aero/Def.	High	19-20%	TS16949, AS9100
Sona BLW	₹3,546 Cr	EV Precision Gears	High	25-27%	IATF 16949
Sansera Engineering	₹3,017 Cr	Auto / Aerospace Machining	High	17-18%	IATF 16949, AS9100
MTAR Technologies	₹676 Cr	Space / Nuclear / Clean Energy	Very High	21-25%	AS9100, NADCAP
Dynamatic Technologies	₹1,404 Cr	Aerospace / UAV / Auto	Very High	18-22%	NADCAP, Boeing-approved
Azad Engineering	₹453 Cr	Aero-Engine Components	Very High	33-36%	NADCAP (multiple)
Data Patterns (India)	₹708 Cr	Defence Electronics	High	35-40%	Mil-Spec, DRDO
Sundram Fasteners	₹5,955 Cr	Fasteners / Shafts	Medium	17-18%	TS16949
Endurance Technologies	₹11,561 Cr	Auto Precision — Brakes/Wheels	Med-High	13-14%	IATF 16949
Ramakrishna Forgings	₹4,034 Cr	Auto / Rail / Power Forgings	Medium	20-22%	TS16949

★ Highlighted rows = high-complexity aerospace/defence specialists with NADCAP-level certification moats and programme-embedded revenue visibility

Complexity vs. Revenue Scale: Competitive Landscape Matrix

		REVENUE SCALE →		
		Small-Mid Revenue	Mid Revenue	Large Revenue
TECHNOLOGY COMPLEXITY	Very High Aerospace / Defence / Nuclear	Azad Engineering Multi-NADCAP · aero-engine	MTAR Technologies Space & nuclear · ISRO lifecycle	Dynumatic Technologies Boeing/HAL · 20-40yr programmes
	High Advanced Machined Components	Unimech Aerospace Aerospace tooling & pneumatics	Sansera Engg. Data Patterns Auto+aerospace · defence electronics	Bharat Forge Global forging leader · A&D pivot
	Medium Precision Auto & Industry	MM Forgings Auto forgings · export-oriented	Ramakrishna Forgings NRB Bearings Rail/power/auto · specialty bearings	Sona BLW Precision EV gears & motors · global OEM mix
	Lower Mass-Produced Parts	LPS Bossard Fasteners · import substitution	Sundram Fasteners Fasteners · bearing caps · shafts	Endurance Technologies Auto brakes & wheels · integrated

Risks & Investment Thesis

Risk framework, three portfolio buckets, due diligence metrics, and the 10-year outlook

Industry Risk Framework: Probability, Severity & Mitigation

Risk	Category	Probability	Severity	Mitigation / Investor Response
Automotive sector slowdown / EV transition disruption	Macro / Demand	Medium	High	Diversify portfolio toward A&D and electronics, avoid ICE-only auto suppliers
Defence budget compression or programme delays	Macro / Demand	Low	High	Multi-programme exposure, prefer export-oriented suppliers
Persistent skills shortage: CNC, GD&T, 5-axis	Structural	High	Medium	Favour companies with structured in-house training academies
Specialty alloy import dependency titanium, Inconel	Structural	Medium	Medium	Long-term supply contracts, backward integration (long-term opportunity)
Infrastructure bottlenecks: power, land acquisition	Structural	Medium	Medium	Prefer companies in established industrial clusters and SEZs
China regaining precision competitiveness post-tariffs	Competitive	Low	High	Certification moat + OEM relationship depth insulates qualified suppliers
Tier-3 fragmentation and price wars	Competitive	High	Medium	Invest at Tier-1/Tier-2 with qualification moats, avoid commodity machining
Programme qualification failure / FAI rejection	Execution	Medium	High	Assess programme portfolio diversity, check quality investment levels
Over-leverage from up-cycle capex expansion	Execution	Medium	High	Monitor Net Debt/EBITDA < 2.5x, prefer FCF conversion > 50%
Additive manufacturing displacing machining niches	Technology	Low (5-10yr)	Medium	Favour companies adopting hybrid additive + subtractive capability
Valuation risk: quality premium over earnings reality	Investor	Medium	High	Focus on earnings quality: order book, ROCE, programme stage

Investment Framework: Three Portfolio Buckets

India's precision engineering sector deserves a structural overweight for 5-10 Year portfolios.

Structured exposure across three conviction tiers, each with distinct risk/return, holding period, and catalyst profile.

01

Long-Cycle Qualified Aerospace & Defence Suppliers

Core Holdings · Highest Conviction

- EBITDA margins 18-30%+
- Order book > 4x annual revenue
- ROCE > 18%; capital productive
- Programme diversity across 3+ OEMs
- Multiple NADCAP special-process accreditations

MTAR Technologies · Dynamatic Technologies ·
Azad Engineering

02

Value Migration Plays — Moving Up the Complexity Curve

Strategic Positions · Thesis Execution Required

- Automotive base provides cash flow
- A&D revenue <20% but growing 40%+ YoY
- Active 5-axis machining capex underway
- First aerospace PPAP / programme approvals secured
- Management explicitly targeting A&D mix shift

Sansera Engineering · Ramakrishna Forgings ·
NRB Bearings

03

EV Transition — New Precision Demand Pool

Thematic Positions · Structural Growth Visible

- EV revenue growing 40%+ YoY
- Qualified across 3+ global OEM EV platforms
- Content-per-vehicle expanding with EV mix
- Low competitive intensity: few qualified suppliers
- India EV exports growing structurally

Sona BLW Precision Forgings · Endurance
Technologies

Key Financial & Operational Metrics Framework

What to track – and what to avoid – when evaluating precision engineering companies for investment.

Metric / KPI	What It Measures	Red Flag ⚠	Green Flag ✓
A&D Revenue as % of Total	Value chain position and margin quality	<10% and not growing	30%+ and growing; explicit A&D strategy
Order Book / Revenue Ratio	Revenue visibility and programme quality	<2x: no multi-year visibility	4x+ with programme-level breakdown
EBITDA Margin %	Complexity level and OEM pricing power	<12%: commodity machining	20%+: certified, programme-embedded
ROCE %	Asset productivity and capital discipline	<12%: capital misallocation	18%+: high-quality, utilisation-driven
Certification Depth	Moat depth and programme eligibility	ISO 9001 only: no premium	AS9100D + multiple NADCAP accreditations
Customer Concentration	Programme concentration risk	>50% single customer: binary risk	<25% top customer; 3+ OEM diversity
Net Debt / EBITDA	Balance sheet quality through capex cycle	>3x: over-leveraged for capex	<1.5x: conservative; FCF-funded growth
Machine Utilisation Rate	Operational efficiency and margin leverage	<55%: poor asset deployment	75%+: operating leverage kicking in
FCF Conversion (FCF/EBITDA)	Capex discipline and earnings cash quality	<35%: heavy capex; earnings not cash	50%+: high cash quality; self-funding
Aerospace/Defence EBITDA Mix	Profitability quality, not just revenue mix	<15% of EBITDA from A&D	35%+ of EBITDA from certified programmes

Sector Outlook: 3, 5 & 10-Year Investment Horizon

3-Year

FY2026-FY2029

Defence indigenisation ramp & first-wave aerospace supply-chain deepening

- ₹3 lakh crore defence production target. Private sector share rising from 21% → 35%+. 16,000 MSMEs upgrading capability.
- Boeing/Airbus sourcing accelerates. First wave of NADCAP-certified Indian shops emerging as global suppliers.
- EV precision component localisation begins at scale. Content-per-vehicle expanding with drivetrain electrification.
- Revenue growth: 20-30% CAGR for well-positioned Tier-1/2 companies. EBITDA margin expansion as mix improves.
- Aerospace specialists reach 40-60x P/E as earnings quality becomes fully understood by capital markets.

5-Year

FY2026-FY2031

India emerges as Tier-1 global precision manufacturing export location

- \$5B+ precision component exports, comparable to Mexico or Taiwan in aerospace manufacturing depth.
- >50% of HAL Tejas Mk2 and AMCA components domestically sourced. New MRFA fighter programme orders begin.
- India's semiconductor mission creates domestic fab-tool precision component market (potential \$2-3B addressable).
- Listed sector market cap expands as earnings quality is fully re-rated by institutional capital.
- International M&A begins: global aerospace Tier-1s acquire Indian qualified shops to embed supply chains.

10-Year

FY2026-FY2036

The IT services parallel — India as a global precision engineering hub

- India precision exports \$15-20B. Global Tier-1 supplier alongside Germany, France, and Taiwan.
- Domestic companies hold AS9100D + NADCAP at scale. Programme portfolios diversified across global OEMs.
- Analogy: India IT services in FY2002 → compounded at 25%+ CAGR for 15 years from similar structural inflection.
- Qualification moat means early investors benefit from permanent competitive advantages, not cyclical positioning.
- 10-20x value creation window for early investors in the right companies.



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India Precision Engineering

Sector Thesis & Deep-Dive

March 2026

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